

$\hat{W}^{EI}$ sets the cliff; $\hat{W}^{IE}$ shapes the basin

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The trained networks this entry uses are produced once in the shared training hub, nb022 (Training), and reused here rather than retrained.

Abstract

Maps the (wEI , wIE) coupling plane: which weight controls the recruitment cliff, and where do *trained* networks land when initialised across it? wEI moves the cliff (E→I drive is the recruitment knob); wIE shapes the basin above engagement but does not move the cliff. Training from each grid point lands in two distinct basins — a low-E PING corner and a high-E silent-I stretched-COBA corner.

Method

Parameter	Value
Integration timestep Δt	0.1 ms
Trial duration T	200 ms
MNIST samples (80/20 stratified split of 2000)	1600 train / 400 test ($\approx 2.9\%$ of the 70k-sample MNIST corpus)
Epochs	10

PING and COBA baseline definitions, training recipe, and the spike-budget regulariser are in nb025. The recruitment cliff at f^* is introduced in nb025. This entry asks: which part of the E↔I coupling architecture controls the cliff’s position, and what happens when we train the network from coupling initialisations spanning both sides of it? Three experiments.

Coupling sweep (inference-time). If f^* is set by the E↔I coupling, altering W^{EI} or W^{IE} at inference should move it. On the trained PING baseline ($\theta_u = \text{off}$), multiply either W^{EI} or W^{IE} by a scalar in $\{0.25, 0.5, 1.0, 2.0\}$ and re-run the W_{in} scale sweep on top. All other weights frozen.

Training grid. A stronger test is to train one network per grid point with W^{EI} and W^{IE} initialised to specified values from scratch, under heavy spike penalty: do the resulting solutions cluster into a PING-like region and a COBA-like region? 5×5 grid over $W^{EI}, W^{IE} \in \{0, 0.25, 0.5, 1.0, 2.0\}$ (mean initialisation, std fixed at $0.1 \times \text{mean}$). One PING-architecture training run per cell at $\theta_u = 0.2$ from epoch 0, $W_{\text{in}} = 0.6$ (a compromise between COBA’s 0.3 init and PING’s 1.2 — gives the no-loop cells a fair shot at converging), **100 training epochs** per cell, seed 42. All other hyperparameters from the standard PING recipe. 25 trainings on Modal A100 in parallel. Reporting per-cell best-epoch accuracy.

Diagonal slice. The grid varies W^{EI} and W^{IE} independently, but the codebase’s standard PING recipe ties them via $W^{IE} = \text{ei_ratio} \cdot W^{EI}$ with $\text{ei_ratio} = 2$. Restricting to that line gives a finer-grained view: one knob (W^{EI}) at 10 values $\{0, 0.1, 0.25, 0.4, 0.5, 0.6, 0.75, 1.0, 1.5, 2.0\}$, W^{IE} tracking at twice the value, **three seeds (42, 43, 44) per value, 100 training epochs**. Best-epoch accuracy reported. Same recipe as the grid otherwise ($\theta_u = 0.2$, $W_{\text{in}} = 0.6$).

Results

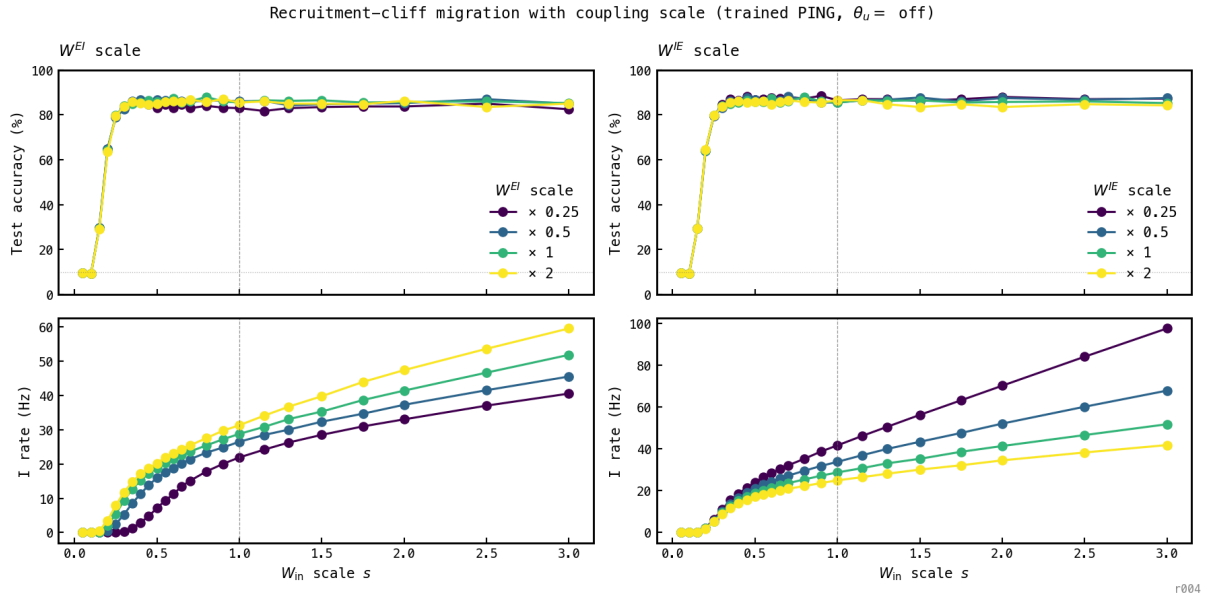


Figure 94: Inference-time W_{in} scale sweep on trained PING ($\theta_u = \text{off}$) with either W^{EI} or W^{IE} scaled. Top: accuracy. Bottom: I rate. Left: W^{EI} sweep, W^{IE} fixed. Right: W^{IE} sweep, W^{EI} fixed.

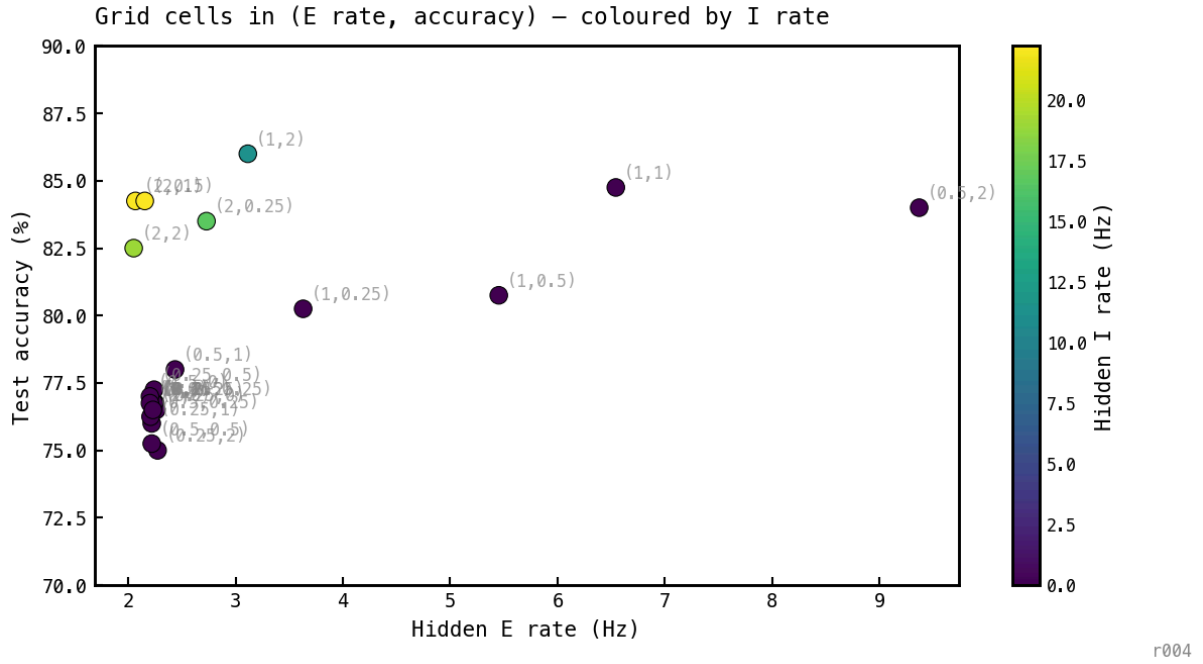


Figure 95: Each dot is one cell of the 5×5 (W^{EI} , W^{IE}) training grid, plotted at its best-epoch test accuracy vs final hidden-E rate. Colour encodes the third quantity — hidden-I rate — which is what discriminates the clusters. Labels are (W^{EI} , W^{IE}).

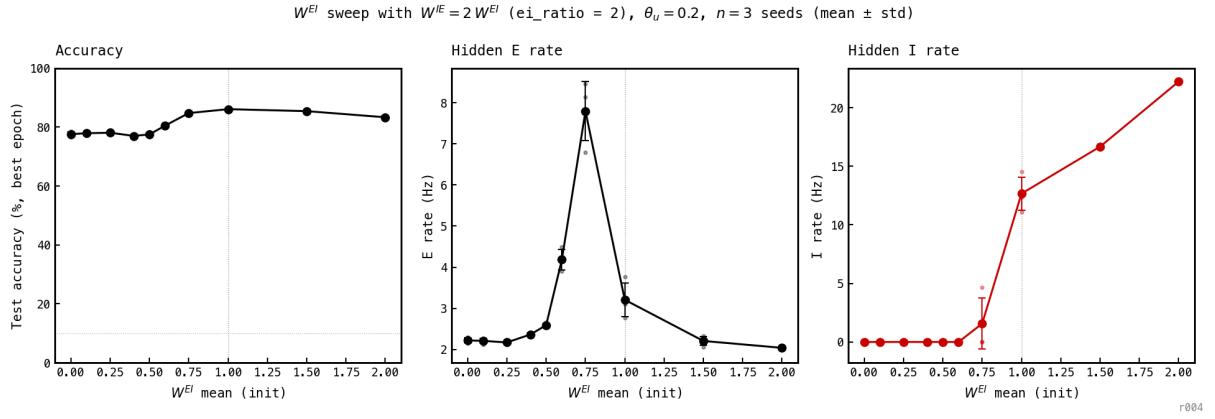


Figure 96: W^{EI} scanned over $\{0, 0.1, 0.25, 0.4, 0.5, 0.6, 0.75, 1.0, 1.5, 2.0\}$ with $W^{IE} = 2W^{EI}$. Error bars are mean $\pm$ std of the best-epoch test accuracy over 3 seeds (42, 43, 44), 100 training epochs each; faint dots show individual seed peaks. **Left:** test accuracy. **Middle:** hidden-E rate (last-epoch). **Right:** hidden-I rate (last-epoch). Grey dotted vertical at $W^{EI} = 1$ marks the canonical recipe baseline.